

INTERPOLACIÓN NUMÉRICA

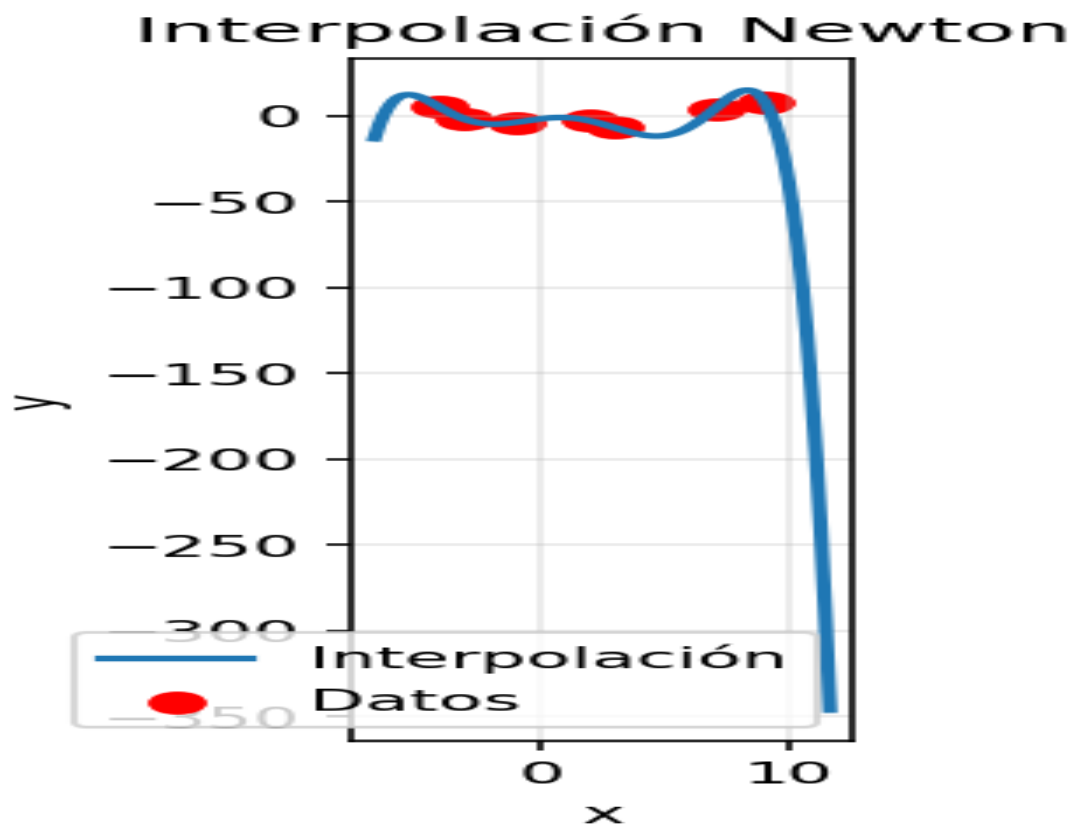
Ejercicio: Interpolación con Newton

Método:	Newton
Iteraciones utilizadas:	21
Punto evaluado:	No especificado
Resultado de la evaluación:	No evaluado
Polinomio interpolante:	$P(x) = -3.0 + 0.3333333333333333 * \prod(x - x_0 \text{ for } j=0 \text{ to } 0) + 0.2666666666666666$

Puntos originales

x	y
2	-3
-1	-4
-3	-2
-4	5
3	-7
7	4
9	8

Gráfica de interpolación



Desarrollo del método

Iteración 1:

ITERACIÓN 1: Diferencia dividida de orden 1

Calculando $f[x_0, x_1] = (f[x_1, x_0] - f[x_0, x_0]) / (x_1 - x_0)$

$$f[x_1, x_0] = -4.0$$

$$f[x_0, x_0] = -3.0$$

$$x_1 = -1.0$$

$$x_0 = 2.0$$

$$f[x_0, x_1] = (-4.0 - -3.0) / (-1.0 - 2.0) = 0.3333333333333333$$

Iteración 2:

ITERACIÓN 2: Diferencia dividida de orden 1

Calculando $f[x_1, x_2] = (f[x_2, x_1] - f[x_1, x_1]) / (x_2 - x_1)$

$$f[x_2, x_1] = -2.0$$

$$f[x_1, x_1] = -4.0$$

$$x_2 = -3.0$$

$$x_1 = -1.0$$

$$f[x_1, x_2] = (-2.0 - -4.0) / (-3.0 - -1.0) = -1.0$$

Iteración 3:

ITERACIÓN 3: Diferencia dividida de orden 1

Calculando $f[x_2, x_3] = (f[x_3, x_2] - f[x_2, x_2]) / (x_3 - x_2)$

$$f[x_3, x_2] = 5.0$$

$$f[x_2, x_2] = -2.0$$

$$x_3 = -4.0$$

$$x_2 = -3.0$$

$$f[x_2, x_3] = (5.0 - -2.0) / (-4.0 - -3.0) = -7.0$$

Iteración 4:

ITERACIÓN 4: Diferencia dividida de orden 1

$$\text{Calculando } f[x_3, x_4] = (f[x_4, x_3] - f[x_3, x_3]) / (x_4 - x_3)$$

$$f[x_4, x_3] = -7.0$$

$$f[x_3, x_3] = 5.0$$

$$x_4 = 3.0$$

$$x_3 = -4.0$$

$$f[x_3, x_4] = (-7.0 - 5.0) / (3.0 - -4.0) = -1.7142857142857142$$

Iteración 5:

ITERACIÓN 5: Diferencia dividida de orden 1

$$\text{Calculando } f[x_4, x_5] = (f[x_5, x_4] - f[x_4, x_4]) / (x_5 - x_4)$$

$$f[x_5, x_4] = 4.0$$

$$f[x_4, x_4] = -7.0$$

$$x_5 = 7.0$$

$$x_4 = 3.0$$

$$f[x_4, x_5] = (4.0 - -7.0) / (7.0 - 3.0) = 2.75$$

Iteración 6:

ITERACIÓN 6: Diferencia dividida de orden 1

Calculando $f[x_5, x_6] = (f[x_6, x_5] - f[x_5, x_5]) / (x_6 - x_5)$

$$f[x_6, x_5] = 8.0$$

$$f[x_5, x_5] = 4.0$$

$$x_6 = 9.0$$

$$x_5 = 7.0$$

$$f[x_5, x_6] = (8.0 - 4.0) / (9.0 - 7.0) = 2.0$$

Iteración 7:

ITERACIÓN 7: Diferencia dividida de orden 2

Calculando $f[x_0, x_2] = (f[x_1, x_1] - f[x_0, x_1]) / (x_2 - x_0)$

$$f[x_1, x_1] = -1.0$$

$$f[x_0, x_1] = 0.3333333333333333$$

$$x_2 = -3.0$$

$$x_0 = 2.0$$

$$f[x_0, x_2] = (-1.0 - 0.3333333333333333) / (-3.0 - 2.0) = 0.26666666666666666$$

Iteración 8:

ITERACIÓN 8: Diferencia dividida de orden 2

Calculando $f[x_1, x_3] = (f[x_2, x_2] - f[x_1, x_2]) / (x_3 - x_1)$

$$f[x_2, x_2] = -7.0$$

$$f[x_1, x_2] = -1.0$$

$$x_3 = -4.0$$

$$x_1 = -1.0$$

$$f[x_1, x_3] = (-7.0 - -1.0) / (-4.0 - -1.0) = 2.0$$

Iteración 9:

ITERACIÓN 9: Diferencia dividida de orden 2

$$\text{Calculando } f[x_2, x_4] = (f[x_3, x_3] - f[x_2, x_3]) / (x_4 - x_2)$$

$$f[x_3, x_3] = -1.7142857142857142$$

$$f[x_2, x_3] = -7.0$$

$$x_4 = 3.0$$

$$x_2 = -3.0$$

$$f[x_2, x_4] = (-1.7142857142857142 - -7.0) / (3.0 - -3.0) = 0.8809523809523809$$

Iteración 10:

ITERACIÓN 10: Diferencia dividida de orden 2

$$\text{Calculando } f[x_3, x_5] = (f[x_4, x_4] - f[x_3, x_4]) / (x_5 - x_3)$$

$$f[x_4, x_4] = 2.75$$

$$f[x_3, x_4] = -1.7142857142857142$$

$$x_5 = 7.0$$

$$x_3 = -4.0$$

$$f[x_3, x_5] = (2.75 - -1.7142857142857142) / (7.0 - -4.0) = 0.40584415584415584$$

Iteración 11:

ITERACIÓN 11: Diferencia dividida de orden 2

$$\text{Calculando } f[x_4, x_6] = (f[x_5, x_5] - f[x_4, x_5]) / (x_6 - x_4)$$

$f[x_5, x_5] = 2.0$
 $f[x_4, x_5] = 2.75$
 $x_6 = 9.0$
 $x_4 = 3.0$

$$f[x_4, x_6] = (2.0 - 2.75) / (9.0 - 3.0) = -0.125$$

Iteración 12:

ITERACIÓN 12: Diferencia dividida de orden 3

Calculando $f[x_0, x_3] = (f[x_1, x_2] - f[x_0, x_2]) / (x_3 - x_0)$

$f[x_1, x_2] = 2.0$
 $f[x_0, x_2] = 0.26666666666666666$
 $x_3 = -4.0$
 $x_0 = 2.0$

$$f[x_0, x_3] = (2.0 - 0.26666666666666666) / (-4.0 - 2.0) = -0.28888888888888889$$

Iteración 13:

ITERACIÓN 13: Diferencia dividida de orden 3

Calculando $f[x_1, x_4] = (f[x_2, x_3] - f[x_1, x_3]) / (x_4 - x_1)$

$f[x_2, x_3] = 0.8809523809523809$
 $f[x_1, x_3] = 2.0$
 $x_4 = 3.0$
 $x_1 = -1.0$

$$f[x_1, x_4] = (0.8809523809523809 - 2.0) / (3.0 - -1.0) = -0.27976190476190477$$

Iteración 14:

ITERACIÓN 14: Diferencia dividida de orden 3

Calculando $f[x2, x5] = (f[x3, x4] - f[x2, x4]) / (x5 - x2)$

$f[x3, x4] = 0.40584415584415584$

$f[x2, x4] = 0.8809523809523809$

$x5 = 7.0$

$x2 = -3.0$

$f[x2, x5] = (0.40584415584415584 - 0.8809523809523809) / (7.0 - -3.0) = -0.04751082251082251$

Iteración 15:

ITERACIÓN 15: Diferencia dividida de orden 3

Calculando $f[x3, x6] = (f[x4, x5] - f[x3, x5]) / (x6 - x3)$

$f[x4, x5] = -0.125$

$f[x3, x5] = 0.40584415584415584$

$x6 = 9.0$

$x3 = -4.0$

$f[x3, x6] = (-0.125 - 0.40584415584415584) / (9.0 - -4.0) = -0.04083416583416584$

Iteración 16:

ITERACIÓN 16: Diferencia dividida de orden 4

Calculando $f[x0, x4] = (f[x1, x3] - f[x0, x3]) / (x4 - x0)$

$f[x1, x3] = -0.27976190476190477$

$f[x0, x3] = -0.28888888888888889$

$x4 = 3.0$

$x0 = 2.0$

$$f[x_0, x_4] = (-0.27976190476190477 - -0.28888888888888889) / (3.0 - 2.0) = 0.00912698412698415$$

Iteración 17:

ITERACIÓN 17: Diferencia dividida de orden 4

Calculando $f[x_1, x_5] = (f[x_2, x_4] - f[x_1, x_4]) / (x_5 - x_1)$

$$f[x_2, x_4] = -0.04751082251082251$$

$$f[x_1, x_4] = -0.27976190476190477$$

$$x_5 = 7.0$$

$$x_1 = -1.0$$

$$f[x_1, x_5] = (-0.04751082251082251 - -0.27976190476190477) / (7.0 - -1.0) = 0.029031385281385283$$

Iteración 18:

ITERACIÓN 18: Diferencia dividida de orden 4

Calculando $f[x_2, x_6] = (f[x_3, x_5] - f[x_2, x_5]) / (x_6 - x_2)$

$$f[x_3, x_5] = -0.04083416583416584$$

$$f[x_2, x_5] = -0.04751082251082251$$

$$x_6 = 9.0$$

$$x_2 = -3.0$$

$$f[x_2, x_6] = (-0.04083416583416584 - -0.04751082251082251) / (9.0 - -3.0) = 0.0005563880563880559$$

Iteración 19:

ITERACIÓN 19: Diferencia dividida de orden 5

Calculando $f[x_0, x_5] = (f[x_1, x_4] - f[x_0, x_4]) / (x_5 - x_0)$

$$f[x_1, x_4] = 0.029031385281385283$$

$$f[x_0, x_4] = 0.00912698412698415$$

$$x_5 = 7.0$$

$$x_0 = 2.0$$

$$f[x_0, x_5] = (0.029031385281385283 - 0.00912698412698415) / (7.0 - 2.0) = 0.003980880230880226$$

Iteración 20:

ITERACIÓN 20: Diferencia dividida de orden 5

Calculando $f[x_1, x_6] = (f[x_2, x_5] - f[x_1, x_5]) / (x_6 - x_1)$

$$f[x_2, x_5] = 0.0005563880563880559$$

$$f[x_1, x_5] = 0.029031385281385283$$

$$x_6 = 9.0$$

$$x_1 = -1.0$$

$$f[x_1, x_6] = (0.0005563880563880559 - 0.029031385281385283) / (9.0 - -1.0) = -0.0028474997224997227$$

Iteración 21:

ITERACIÓN 21: Diferencia dividida de orden 6

Calculando $f[x_0, x_6] = (f[x_1, x_5] - f[x_0, x_5]) / (x_6 - x_0)$

$$f[x_1, x_5] = -0.0028474997224997227$$

$$f[x_0, x_5] = 0.003980880230880226$$

$$x_6 = 9.0$$

$$x_0 = 2.0$$

$$f[x_0, x_6] = (-0.0028474997224997227 - 0.003980880230880226) / (9.0 - 2.0) = -0.0009754828504828498$$